homework_1

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1 Homework 1

DATA 37711 — written answers to Problems 1–3.

1.1 Problem 1

Setup. In the 2020 US Presidential election, poll aggregators (e.g., fivethirtyeight.com) and betting markets disagreed about the probability of a Biden victory. Under Bayesian decision theory, a rational agent writes down a probability distribution $P(X)$ over states of the world and a utility function $U(a, X)$, then chooses

$$a^* = \arg\max_a \ \mathbb{E}_{X\sim P}\big[U(a, X)\big].$$

Victor is a rational agent with d dollars of total wealth. He believes $P(\text{Biden wins}) = 0.85$ (the fivethirtyeight prediction), and betting markets give even-money bets at $P(\text{Biden wins}) = 0.6$.

1.1.1 (a)

Question. Prove that if Victor's utility function is just the amount of money he has, he'll bet d (his entire bankroll) on Biden winning. Does linear utility of money make sense?

Answer.

Payout. The market odds are fair for someone who believes $P(\text{Biden wins}) = 0.6$, so a bet of a dollars has zero expected profit under that belief:

$$0.6 \cdot \text{profit} - 0.4\, a = 0 \quad \Longrightarrow \quad \text{profit} = \tfrac{2}{3}\, a.$$

So each dollar bet on Biden earns 2/3 of a dollar if Biden wins.

Utility. Let $a \in [0, d]$ be the amount bet on Biden. Victor's final wealth is

$$U(a, \text{Biden wins}) = d + \tfrac{2}{3}a \quad \text{(he gets his stake back plus the profit)},$$
$$U(a, \text{Biden loses}) = d - a.$$

Expected utility under Victor's beliefs:

$$\mathbb{E}[U(a, X)] = 0.85\,\big(d + \tfrac{2}{3}a\big) + 0.15\,(d - a) = d + \big(0.85 \cdot \tfrac{2}{3} - 0.15\big)\, a = d + \tfrac{5}{12}\, a.$$

The feasible actions are $0 \le a \le d$, since he can't bet more than he has. The coefficient $5/12$ is strictly positive, so $\mathbb{E}[U]$ is strictly increasing in a, and its maximum on $[0, d]$ is at the endpoint $a^* = d$. Victor bets his entire bankroll. ■

(The coefficient is positive because Victor's belief, 0.85, exceeds the market's, 0.6; if they were equal, it would be zero and he would be indifferent.)

Does linear utility make sense? No. Under this strategy, Victor has a 15% chance of losing everything. People generally care more about avoiding ruin than about gaining an equal amount of money (risk aversion), so betting everything on one event is not reasonable. Also, the value of an extra dollar depends on how much you already have: a fixed gain means less to someone with large d than to someone with small d (diminishing marginal utility). Both ideas suggest utility should be a concave, nonlinear function of total wealth. Under such a utility, a reasonable agent with an edge would still bet something, but not everything.

1.1.2 (b)

Question. Now suppose Victor's utility from y dollars is $U(y) = \log(y)$. How much (of his d dollars) should he bet on Biden winning?

Answer.

Final wealth in each outcome is the same as in (a), so

$$U(a, \text{Biden wins}) = \log\!\left(d + \tfrac{2}{3}a\right), \qquad U(a, \text{Biden loses}) = \log(d - a).$$

Expected utility:

$$E(a) = 0.85 \log\!\left(d + \tfrac{2}{3}a\right) + 0.15 \log(d - a), \qquad 0 \le a < d.$$

First-order condition:

$$\frac{dE}{da} = \frac{0.85 \cdot \frac{2}{3}}{d + \frac{2}{3}a} - \frac{0.15}{d - a} = 0 \implies 0.85 \cdot \tfrac{2}{3}\,(d - a) = 0.15\left(d + \tfrac{2}{3}a\right).$$

Multiplying by 3:

$$1.7\,(d - a) = 0.45\,d + 0.3\,a \implies 1.25\,d = 2a \implies a^* = 0.625\,d = \tfrac{5}{8}\,d.$$

This is a maximum: log is concave, so each term of E is concave in a, which makes E concave and the critical point its global maximum on $[0, d)$.

So Victor bets 5/8 of his wealth. This is between 0 and d and less than the "bet everything" answer from (a): log utility makes him keep some money in reserve, but his edge still leads to a large bet. This matches the Kelly criterion, $f^* = p - (1 - p)/b$ with $p = 0.85$ and $b = 2/3$:

$$0.85 - \frac{0.15}{2/3} = 0.625.$$

[1]:
```
import numpy as np

# Numerical check of 1(b) with d = 1: expected log utility on a fine grid of a␣
↪in [0, d)
d = 1.0
a = np.linspace(0, d, 1_000_001, endpoint=False)  # a in [0, d)
expected_log_utility = 0.85 * np.log(d + (2 / 3) * a) + 0.15 * np.log(d - a)

a_star = a[np.argmax(expected_log_utility)]
print(f"grid maximizer a* = {a_star:.6f}  (analytic: 5/8 = 0.625)")
assert abs(a_star - 0.625) < 1e-4
```

```
grid maximizer a* = 0.625000  (analytic: 5/8 = 0.625)
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1.1.3 (c)

Question. Suppose you had a net worth of $500,000. Would you personally make the bet from part (b)?

Answer (my opinion).

No, not at that size. 5/8 of $500,000 is $312,500, which means a 15% chance of losing more than $300,000 on a single event. Kelly is optimal only if the model's assumptions hold, and I don't think they fully apply to me:

- The 0.85 is an estimate, not a known fact. The market's 0.6 is also information, and if my probability is too high, full Kelly overbets, which is penalized much more than underbetting. That's why practitioners often use fractional Kelly (e.g., half Kelly).
- My personal utility is likely more risk-averse (more concave) than log, and under a more concave utility the optimal bet really is smaller.
- The model assumes this bet is my only option. In reality I'd rather keep most of my money in more reliable, diversified investments.

I would still bet on Biden if I held that conviction, but only a small percentage of my wealth.

1.2 Problem 2

Question. Suppose Y is a categorical random variable and X are covariates used to predict Y. A loss function L on categorical data is a *proper scoring rule* if the predictive function that minimizes the implied risk is the conditional probability of Y given X; that is, if

$$f^* := \arg\min_f \mathbb{E}_{X,Y}[L(Y, f(X))] \quad \text{is given by} \quad f^*(x) = P(Y \mid X = x).$$

Prove that the cross-entropy loss is a proper scoring rule when Y is binary.

Answer.

For binary $Y \in \{0, 1\}$, the prediction $f(x) = q \in (0, 1)$ is the predicted probability that $Y = 1$. The binary cross-entropy loss is

$$L(y, q) = -\big[\, y \log q + (1 - y) \log(1 - q) \,\big],$$

which is $-\log q$ when $y = 1$ and $-\log(1 - q)$ when $y = 0$.

Step 1 (law of total expectation).

$$\mathbb{E}_{X,Y}\big[L(Y, f(X))\big] = \mathbb{E}_X\Big[\, \mathbb{E}_{Y|X}\big[L(Y, f(X)) \mid X\big]\Big].$$

Step 2 (inner expectation). Fix x, and let $p = P(Y = 1 \mid X = x)$ and $q = f(x)$. Then

$$\mathbb{E}_{Y|X}\big[L(Y, q) \mid X = x\big] = -\big[\, p \log q + (1 - p) \log(1 - q) \,\big] =: g(q).$$

Step 3 (minimize over q).

$$g'(q) = -\frac{p}{q} + \frac{1 - p}{1 - q} = 0 \implies (1 - p)\, q = p\, (1 - q) \implies q = p.$$

Moreover,

$$g''(q) = \frac{p}{q^2} + \frac{1 - p}{(1 - q)^2} > 0 \quad \text{on } (0, 1),$$

so g is strictly convex and $q = p$ is its unique minimizer.

Step 4 (pointwise minimization suffices). Because f is an arbitrary function, its value at each x can be chosen independently of its value at any other x. For every x, the inner expectation is at least its minimum value, which is achieved at $f(x) = P(Y = 1 \mid X = x)$. Since expectation preserves inequalities (if $g(x) \geq h(x)$ for all x, then $\mathbb{E}[g(X)] \geq \mathbb{E}[h(X)]$), the outer expectation is at least the expectation of the pointwise minimum, and choosing $f(x) = P(Y = 1 \mid X = x)$ at every x achieves that bound.

Therefore $f^*(x) = P(Y = 1 \mid X = x)$. For binary Y this determines the whole conditional distribution, since $P(Y = 0 \mid X = x) = 1 - P(Y = 1 \mid X = x)$, so $f^*(x) = P(Y \mid X = x)$ and cross-entropy is a proper scoring rule. ■

1.3 Problem 3

Question. As we saw in class, a requirement for stochastic gradient descent to converge to the minimum of an objective function is that the step size ρ_t decays fast enough that $\sum_t \rho_t^2 < \infty$. In practice it's common to use a constant step size, violating this requirement. Why might this be reasonable in a finite sample setting?

Answer.

The condition $\sum_t \rho_t^2 < \infty$ guarantees that SGD converges to the exact minimizer of the objective as $t \to \infty$. In a finite-sample setting, though, the objective we optimize is the empirical risk, and its minimizer is not the minimizer of the true risk we care about. Because we only have n samples, the empirical minimizer already differs from the true-risk minimizer by a statistical error that doesn't vanish for fixed n (typically on the order of $1/\sqrt{n}$).

With a constant step size ρ, SGD doesn't converge to a single point. Because the gradient noise never shrinks, it settles into a neighborhood of the empirical minimizer whose size depends on ρ. If ρ is chosen so that this neighborhood is small compared to the statistical error, then the extra precision from decaying the step size toward zero barely improves performance on new data. The total error is dominated by the finite sample, not by the optimization.

A constant step size also has practical advantages. Decaying step sizes eventually become very small, so progress slows down a lot, while a constant step reaches the "good enough" region much faster. And since exactly minimizing the empirical risk can lead to overfitting, the leftover noise from a constant step (often combined with early stopping) can act as a form of implicit regularization.

1.4 Coding problems

PLACEHOLDER — TO BE ADDED. The remaining (coding) problems for Homework 1 will be added in this section.